

Natural Language Processing

Expanded Knowledge Base Document for RAG / Document Question-Answering

1. Introduction

Natural language processing, or NLP, concerns computational methods for working with human language. Common tasks include text classification, sentiment analysis, named entity recognition, question answering, summarization, translation, and information extraction.

2. Text Representation

Text must generally be converted into a representation that a model can process. Traditional approaches include bag-of-words and TF-IDF. Bag-of-words represents documents using word occurrence information, while TF-IDF adjusts word importance based on how frequently a term appears in a document and how common it is across a collection.

3. Embeddings

Neural systems commonly use dense vector representations called embeddings, which encode semantic and contextual information. Embeddings can represent words, sentences, paragraphs, or documents. They are useful for similarity search, classification, clustering, retrieval, and other language-processing tasks.

4. Tokenization

Tokenization converts raw text into model-readable units. Depending on the model, tokens may correspond to words, subwords, or characters. Tokenization is an important preprocessing step because the model operates on token sequences rather than directly on raw human-readable text.

5. Transformers and Attention

Modern language models use transformer architectures based on attention mechanisms. Attention allows a model to consider relationships between tokens when producing a representation or prediction. This makes transformers useful for processing contextual relationships across sequences and supports many modern NLP applications.

6. NLP Tasks

Text classification assigns categories to text, such as topic or sentiment. Named entity recognition identifies entities such as people, organizations, places, or other domain-specific items. Question answering produces an answer to a question. Summarization creates a shorter representation of a longer text, while information extraction converts unstructured text into structured fields.

7. NLP Evaluation

NLP systems can be evaluated using task-specific metrics. Classification tasks may use accuracy or F1. Generation tasks can use several automatic metrics together with human evaluation. Evaluation should reflect the actual purpose of the system because a metric that works for classification may not adequately measure the quality of generated text.

8. NLP and RAG

NLP techniques are closely connected to retrieval-augmented generation. Document parsing, tokenization, embeddings, semantic similarity, and language generation can all appear in a RAG pipeline. In a document question-answering system, NLP methods help transform documents and queries into representations that support retrieval and response generation.

9. Practical Concerns

Important practical concerns include ambiguity, domain-specific vocabulary, bias, privacy, and factual reliability. The same word can have different meanings depending on context. Specialized domains may use terminology that is rare in general text. Systems should therefore be evaluated using representative data and appropriate quality checks.

10. Example NLP Workflow

A simple NLP workflow may begin with collecting text, cleaning or normalizing it when appropriate, tokenizing the text, creating a representation, training or applying a model, and evaluating results. For modern transformer-based applications, the workflow may instead rely heavily on pretrained models and embeddings while still requiring careful data preparation and evaluation.

11. Key Takeaways

NLP provides methods for converting human language into representations that computational systems can process. Modern NLP combines tokenization, embeddings, attention-based architectures, and task-specific evaluation. These techniques also provide important building blocks for document retrieval and question-answering systems.